Supplementary Material for "Auditing Refresh Scheduling in Collaborative Sensing"

Haojie He
School of Artificial Intelligence
Tianjin University of Science and Technology
Tianjin 300457, China
Email: hhj18393132200@mail.tust.edu.cn
ORCID: 0009-0005-7327-6141

I. Overview

This document supplements the main paper. Section II details the experimental setup; Section III reports additional results of the original protocol, including sensitivity checks and participant-specific payloads; Section IV gives the matched offline schedule; Section V summarizes the causal-imitation experiment; Section VI reports the auxiliary self-triggered push, its learned variant, and the two-sided trigger; Section VII documents the cost-aware design-selection experiment; and Section IX describes reproducibility. Code, frozen protocols, and summary tables are available at https://github.com/HeHaojie-Agent/cscwd2027-refresh-audit. Its file names and scripts carry stage labels: V20 for the matched forecaster comparison on PAMAP2 and MHEALTH, V23 for the external OPPORTUNITY replication, V28 for the offline schedule, V29 for causal imitation, V30 for the frozen self-triggered push, V30b for its post hoc frontier extension, and V31 for the learned auxiliary detector, and V32 for the two-sided trigger.

II. Detailed Setup

Two-second windows step by one second before labels are attached. Gaps, records, and invalid windows reset caches. Forward fill is causal and capped at ten PAMAP2 or three OPPORTUNITY samples; MHEALTH uses nominal row timestamps. The three evaluation grids retain every window (stride 1, offset 0, the primary grid), or alternating windows with stride 2 and offset 0/1 within each contiguous stream. The latter grids have nonoverlapping windows; models are fitted separately per grid.

The original caches contain 28,450/24,301 windows, including 9,014/17,184 unknown-only and 362/493 mixed windows. OPPORTUNITY retains 20,995 windows. An endpoint outside the chosen classes is unknown; a mixed window contains known samples but no single known activity throughout. All retained windows enter byte/call denominators, and endpoints with known labels enter F1. Twelve slots are quantized uniformly from $[-4, 4]$ to uint8; fusion sums integer scores, breaking ties by smallest index. OPPORTUNITY maps locomotion labels 1/2/4/5 to four active classes, eight fixed -4 slots, and four-class macro-F1. Primary-grid reference F1 is 0.7360/0.9095/0.6966.

Five monotone families derive from this paid state. Scalar controls see only p from the S current-risk head: geometric extrapolation $1-(1-p)^h$ and horizon-specific sigmoid/isotonic calibration. Full-feature models use direct logistic heads with cumulative maxima, or discrete hazards over $[0]$, $[1, 2]$, $[3, 6]$, $[7, 14]$, giving $F(h_j) = 1-\prod_{\ell \le j}(1-\lambda_\ell)$. Horizons are 1/3/7/15. Logistic fits use standardized inputs, subject-equal weights, $C = 1$, at most 500 iterations, and at most 512 uniformly spaced training landmarks/person; constant targets use a 0.5 pseudocount.

All families run at fixed $\epsilon = 0.01, 0.05, 0.1$, with hazard 0.01 the main policy and Platt 0.01 the main scalar control. Periods 1/2/4/8/16 and both complete-push codecs yield 22 policies. Seven causal event rules add primary-drift thresholds 0.01/0.025/0.05/0.1 and top-two-gap thresholds 4/16/32, all with maximum age 16. Drift divides primary L1 score change by 12×255. A B-only selector chooses the minimum-byte eligible rule, using fixed tie order and complete-push fallback, before opening A scores. Drift 0.025 remains the main event point.

Each deadline or event decision costs two 15-byte messages, including immediate refresh; periodic push has no repeated notices. Lossless score packets use Rice2 temporal residuals and raw escape, referencing the last score actually sent. Configurations cost 134 bytes/session (116 for complete push), sessions last 2,048 windows, and status costs 45 bytes every 100 windows and at resets. Model packages cost 454/1,726/486 bytes for geometric/direct/hazard, 518 for Platt, and $470+16K$ for isotonic's K knots. Complete-push baselines use fixed Rice2 for overlapping PAMAP2 and Rice3 otherwise. Byte counts include all these mutually exclusive charges.

In the development data, evaluation subjects A are excluded from node fitting S and future-head/calibrator fitting B. Each development fold (view) omits its previously designated validation/test subjects. Its sorted remaining six PAMAP2 or eight

MHEALTH subjects are partitioned into consecutive pairs; each pair serves as A, yielding $8 \times 3 = 24$ or $10 \times 4 = 40$ groups. Within the remaining subjects, alternating sorted halves form S and B and then swap roles. Three grids yield $64 \times 2 \times 3 = 384$ cases. B and A share frozen S node models and a current-risk head fitted on subject-out-of-fold S scores. The original datasets had prior development exposure; these repeated views are not a new outer test.

A post hoc influence check removes one subject's evaluation contribution within each dataset/grid and averages the remainder without refitting. The removed subject may retain S/B roles in other views. This tests summary sensitivity, not new subject-held-out validation. Recovery mixtures are not recomputed, so it cannot establish full joint-screen deletion robustness.

III. Additional Results of the Original Protocol

TABLE I

Distinct evidence panels. (a) Single-host raw execution on the original subjects' first 256 windows: subject-mean medians over three rounds, amortized wall ms/window. (b) Failed original-implementation joint-screen gates on full primary-grid streams; B, bytes; S, positive subjects; C, calls; D, disagreement; F, F1; R, recovery. Neither panel pools prefixes with complete streams.

(a) Local execution, ms/window

Policy	PAMAP2	MHEALTH
Complete push	5.0745	3.2741
Period 2	3.4364	2.1674
Platt 0.01	3.8832	2.9370
Hazard 0.01	3.7471	3.1819
Drift 0.025	2.1423	1.8575

(b) Full-stream failed conditions

Policy	PAM	MHEALTH	OPP
Period 2	D	D	D,F,R
Platt 0.01	B,S	B,S	B,S,C
Direct 0.01	B,S	B,S	B,S
Hazard 0.01	B,S,C	B,S,C	B,S,C
Drift 0.025	D,R	D	D,F,R

TABLE II

Executable protocol factorial on the OPPORTUNITY primary grid. Columns report mean-byte ratios to complete push within the same implementation: L/C denote legacy/compact control, 12/4 denote score slots. Calls and disagreement D (%) are identical across implementations. Last column lists failed compact-four gates: B, 10% mean saving; S, 75% subjects saving; C, calls ≤ 2.7; D, disagreement $\leq 1\%$; F, F1 loss ≤ 0.005; R, either recovery condition.

Policy	L12	C12	L4	C4	Calls	D (%)	Failed C4 gates
Complete push	1.000	1.000	1.000	1.000	3.000	0.00	B,S,C
Period 2	0.681	0.681	0.682	0.682	2.002	6.19	D,F,R
Platt 0.01	1.330	1.012	1.434	1.040	2.777	0.01	B,S,C
Isotonic 0.01	1.188	0.920	1.277	0.946	2.495	0.24	B
Direct 0.01	1.298	0.994	1.399	1.023	2.698	0.16	B,S
Hazard 0.01	1.448	1.091	1.567	1.124	2.997	0.01	B,S,C
Drift 0.025	0.604	0.524	0.629	0.530	1.468	10.43	D,F,R
Complete-push B/window	66.33	66.36	53.54	53.54	–	–	–

The protocol factorial (Table II) used calibration on all B records and six-byte paired control, rather than the record-separated calibration and ten-byte pair of the design-selection experiment. It replays 336 complete streams and 112 raw-window paths. Main hazard overhead is 44.79% for legacy twelve slots, 9.07% for compact twelve, 56.68% for legacy four, and 12.44% for compact four. Primary isotonic saving is 7.95% for compact twelve and 5.45% for compact four, below the ten-percent target.

TABLE III

Primary-grid conclusion sensitivity, main hazard 0.01. R is application bytes relative to complete push; individual ranges use each subject, deletion ranges remove one evaluation contribution without refitting. Last column counts deletions where hazard has lower Brier than isotonic.

Dataset	n	Individual R	Deletion R	Lower Brier
PAMAP2	8	1.296–1.407	1.354–1.370	7/8
MHEALTH	10	1.344–1.427	1.385–1.394	6/10
OPPORTUNITY	4	1.442–1.451	1.447–1.450	1/4

TABLE IV
Common-landmark hazard minus isotonic at $\epsilon = 0.01$, primary grid. Same/longer intervals are percentages. $\Delta B_S, \Delta B_C$ are weighted same/changed-action Brier contributions, summing to ΔB.

Dataset	ΔB	Same	Longer	ΔB_S	ΔB_C
PAMAP2	-0.001642	65.41	3.38	-0.001683	+0.000041
MHEALTH	+0.000174	65.23	1.58	-0.002598	+0.002772
OPPORTUNITY	+0.007752	58.62	0.16	+0.005297	+0.002455

Table IV gives the per-dataset decision-region decomposition summarized in Section V-A of the main paper.

The original main-hazard individual byte ratios span 1.296–1.407, 1.344–1.427, and 1.442–1.451 in dataset order. All sixty-six evaluation-summary deletions retain mean-byte increases and uniform-payload exclusion. These leave training roles intact, add no independent subjects, and do not recompute recovery mixtures. Seven of eight PAMAP2 subjects individually satisfy the uniform exclusion. Primary Brier advantages survive seven/eight, six/ten and one/four deletions; PAMAP2 Brier differences range from -0.003300 to +0.000263 and changed-action contributions from -0.000554 to +0.000557.

For the original matched direct predictor, main-grid control-pair budgets are 5.22/4.21 bytes for PAMAP2/MHEALTH. Formal additional-payload floors at thirty-byte control are 70.65/78.92/10,727.98 bytes per packet in dataset order. These are fixed-trace algebraic counterfactuals. Both external nonoverlap grids admit no payload-only repair. For participant-specific additions u_i to node i's score packets, the uniform term uK_η of the main-text envelope becomes $\sum_i u_i[(1-\eta)k_{0,i} - k_i]$, so excluding arbitrary additions requires every coefficient to be nonpositive. Eight of the nine main-hazard dataset–grid settings satisfy this; overlapping PAMAP2 is the exception. The heterogeneous exception has PAMAP2 primary coefficient -0.1 and auxiliary coefficients 0.005236; auxiliary-only floors are 230.30 bytes with free control or 2,784.63 bytes at the implemented charge. The exception still fails the call gate. Original full external main-hazard cost is 96.0366 versus 66.3303 B/window, 2.9969 calls/window and 0.0143% disagreement.

A. Stability of Audit Conclusions

Under the original implementation, main hazard increases primary-grid bytes for every PAMAP2, MHEALTH, and OPPORTUNITY subject. All 66 evaluation-summary deletions preserve the mean-byte increase and uniform-payload exclusion. These sensitivity checks retain training roles and do not recompute recovery mixtures. Predictive ordering varies more: hazard has lower primary-grid Brier after 7/8, 6/10, and 1/4 deletions, respectively. Both the PAMAP2 Brier difference and changed-action contribution reverse sign. The sample-specific forecast decomposition is therefore less stable than the observed direction of communication cost.

Original single-core timing also separates computation from communication. Across five policies and 256 primary-grid windows/person, all 270 interleaved runs match frozen traces. Main hazard reduces prefix wall time by 26.16%/2.82% on PAMAP2/MHEALTH while increasing prefix bytes by 6.83%/44.27%. Table I lists the timings and failed conditions.

IV. Matched Offline Opportunity: Protocol and Complete Results

This post hoc diagnostic uses all 384 original and twelve OPPORTUNITY case–grid combinations, with the same quantized scores and resets as the matched forecast comparisons. Directions and views retain the original subject-equal aggregation. It adds neither raw records nor model fits. Paired control charges are zero and thirty bytes, and score widths are twelve plus four for OPPORTUNITY only. Narrow packing and its reference both add fifty-four schema bytes/session. The zero-control condition is an algebraic idealization, not a deployed transport.

The backward recursion minimizes auxiliary Rice2 packet bytes and per-decision control under zero reference disagreement. Packets reference the last auxiliary score actually sent. Session-initial score packets, primary traffic, status, and original configuration are charged separately. Complete push uses the original codec and its own configuration. A second analysis adds each matched hazard package deployment charge to the offline total. This charge is constant across candidate paths and therefore does not change the optimum. Online planning costs are not estimated. Future scores and segment tails are freely observed; actual model calls remain three/window. No call, radio, or online learning improvement follows from these byte results.

Implementation checks compare forty-eight tiny cases with exhaustive path enumeration and twenty-four full streams with a separately coded forward shortest path. All 1,584 full-stream schedules are checked for zero reference disagreement, valid resets/deadlines and score/control/fixed-fee identities. Deterministically sampled actual encode/decode checks cover every reset and score quantiles; they are not a claim of decoding every packet. The frozen protocol and two source-correction records precede the relevant execution stages. The tiny-case test caught unsigned edge subtraction before any full stream was scored; a later external-baseline correction fixed the original-only helper before any external stream was scored, preserving all 1,536 original traces.

TABLE V: Fee-matched offline mean-byte opportunity, including the matched hazard package. Every path has zero reference disagreement; actual oracle calls remain three/window.

Dataset	Grid	Width	Pair B	Saving (%)	B/window
MHEALTH	1/0	12	0	48.752	33.5698
MHEALTH	1/0	12	30	37.978	40.6275
MHEALTH	2/0	12	0	44.714	36.8835
MHEALTH	2/0	12	30	31.965	45.3893
MHEALTH	2/1	12	0	44.640	36.9364
MHEALTH	2/1	12	30	31.839	45.4771
OPPORTUNITY	1/0	4	0	50.713	26.3867
OPPORTUNITY	1/0	4	30	39.073	32.6180
OPPORTUNITY	1/0	12	0	51.099	32.4365
OPPORTUNITY	1/0	12	30	41.702	38.6694
OPPORTUNITY	2/0	4	0	43.460	31.1134
OPPORTUNITY	2/0	4	30	26.607	40.3877
OPPORTUNITY	2/0	12	0	42.463	40.6344
OPPORTUNITY	2/0	12	30	29.328	49.9112
OPPORTUNITY	2/1	4	0	43.915	30.8591
OPPORTUNITY	2/1	4	30	27.411	39.9396
OPPORTUNITY	2/1	12	0	42.896	40.2651
OPPORTUNITY	2/1	12	30	30.015	49.3475
PAMAP2	1/0	12	0	50.777	31.1430
PAMAP2	1/0	12	30	40.760	37.4804
PAMAP2	2/0	12	0	47.259	34.5198
PAMAP2	2/0	12	30	35.379	42.2959
PAMAP2	2/1	12	0	47.348	34.4587
PAMAP2	2/1	12	30	35.517	42.2016

The offline minimum is exact only within the stated zero-conflict, bounded-deadline family. A zero-conflict feasible path witnesses communication opportunity. Failure of this stricter family would not prove impossibility for policies permitting one-percent disagreement; success likewise does not prove a causal scheduler can find it. The known-schedule transmission counts in the result files are hypothetical and are never reported as actual oracle computation savings.

V. Causal Imitation of the Offline Schedule

Policies are trained on B subjects to imitate the zero-conflict offline choices of Section IV from the 52 paid age-one features (main paper, Section III-E). Table VI lists, for each dataset–grid, the highest byte saving among settings with at most 1% reference disagreement. No imitation setting passes the joint screen in any of the nine scenarios, whether or not the package fee is charged. Labels use future scores of B only; A evaluation followed a frozen protocol (`frozen.json` in the repository). The external cases were also rerun on a second machine with a different scikit-learn version; savings differ by at most 0.64 and disagreement by at most 0.27 percentage points, and the reported values come from the full run on the original machine.

TABLE VI

Best byte saving (%) with reference disagreement (%) in parentheses, among settings with at most 1% disagreement. "Deployable" covers the charged linear and boosted imitators; "diagnostic" adds the fee-free large model; forecasters are the twelve threshold policies; offline is the fee-matched zero-conflict schedule.

Dataset	Grid	Deployable imitation	Incl. diagnostic	Best forecaster	Offline
PAMAP2	all	-19.68 (0.69)	-4.39 (0.96)	-9.03 (0.66)	40.76
PAMAP2	alt 0	-30.73 (0.59)	-30.73 (0.59)	-18.90 (0.60)	35.38
PAMAP2	alt 1	-30.33 (0.60)	-30.33 (0.60)	-17.60 (0.77)	35.52
MHEALTH	all	-14.46 (0.77)	-10.80 (0.48)	-7.42 (0.82)	37.98
MHEALTH	alt 0	-24.79 (0.57)	-18.33 (0.56)	-11.76 (0.99)	31.96
MHEALTH	alt 1	-25.15 (0.94)	-18.65 (0.56)	-11.75 (0.94)	31.84
OPPORTUNITY	all	-15.34 (0.86)	-5.45 (0.93)	-4.92 (0.95)	41.70
OPPORTUNITY	alt 0	-29.94 (0.89)	-21.30 (0.97)	-27.12 (0.79)	29.33
OPPORTUNITY	alt 1	-31.87 (0.59)	-31.87 (0.59)	-31.12 (0.53)	30.02

VI. Auxiliary Self-Triggered Push and Two-Sided Trigger

The causal-imitation result locates the missing timing information outside the primary node's paid state. The self-triggered push of the main paper (Section III-F) moves the trigger to the auxiliaries. Each auxiliary computes the per-axis mean and standard deviation of its own cleaned window, six axes on PAMAP2 and three otherwise, without running its classifier. It pushes a Rice2 score packet when the largest normalized change since its last push exceeds κ, when its cached scores reach age sixteen, or at a reset. The two auxiliaries decide independently; the primary infers every window and fuses its current scores with the latest scores it holds. No notice or request messages are sent. The ledger charges score packets, the 134-byte configuration and 45-byte status messages, and a per-session trigger package carrying float32 noise scales and κ for both auxiliaries (134 bytes on PAMAP2, 86 bytes otherwise).

The noise scale of each descriptor component is fitted on B only, as the subject-equal mean of per-subject median absolute changes between consecutive windows within a segment. The deployable κ is chosen on B from $\{1, 1.5, 2, 3, 4, 6, 8, 12, 16\}$ as the lowest-byte value meeting the B-measurable gates (saving, positive subjects, calls, disagreement, and F1), and otherwise the smallest value. The protocol, including the source hash, was frozen before any A evaluation. A B-only self-test recomputes descriptors directly from raw windows, replays the push rule and cached fusion in an independent implementation, and checks that forcing every push reproduces the complete-push score ledger exactly. All 396 case–grids were evaluated; $\kappa = 1$ was selected in 394 and $\kappa = 1.5$ in two.

After the frozen results were seen, $\kappa \in \{0.5, 0.6, 0.75, 0.9\}$ were added as a declared post hoc frontier extension with the same B-fitted scales, ledger, and screen. These points describe the saving–disagreement trade-off near the 1% limit and are not deployable selections. Table VII lists the frozen selection and the best extended point within 1% disagreement.

TABLE VII

Auxiliary self-triggered push. Frozen: B-selected κ, with byte saving (%), reference disagreement (%), calls/window, subjects with positive saving (%), known-endpoint macro-F1 loss (percentage points), and failed gates (B, bytes; D, disagreement). Last column: highest saving among all fixed κ (frozen and post hoc) with at most 1% disagreement, with disagreement and κ; – means no κ reaches 1%.

Dataset	Grid	Saving	Disagr.	Calls	Pos.	F1 loss	Failed	Best within 1%
PAMAP2	all	11.85	1.69	2.613	100	-0.06	D	8.02 (0.92; $\kappa = 0.5$)
PAMAP2	alt 0	11.39	2.02	2.610	100	-0.10	D	–
PAMAP2	alt 1	11.34	2.16	2.611	100	-0.06	D	–
MHEALTH	all	12.97	1.49	2.561	100	0.03	D	9.73 (0.90; $\kappa = 0.6$)
MHEALTH	alt 0	11.85	1.61	2.558	100	-0.08	D	8.29 (0.92; $\kappa = 0.6$)
MHEALTH	alt 1	11.94	1.75	2.555	100	0.03	D	7.52 (0.90; $\kappa = 0.5$)
OPPORTUNITY	all	11.54	1.08	2.629	100	0.08	D	10.40 (0.95; $\kappa = 0.9$)
OPPORTUNITY	alt 0	9.43	2.12	2.614	100	-0.56	BD	3.34 (0.93; $\kappa = 0.5$)
OPPORTUNITY	alt 1	9.37	2.14	2.615	100	0.16	BD	–

The frozen selection improves every evaluation subject's bytes and keeps calls and F1 within their gates, but its disagreement exceeds 1% in all nine scenarios, and the two OPPORTUNITY nonoverlap grids also miss the 10% saving target. Among extended points only $\kappa = 0.9$ on the OPPORTUNITY primary grid passes the full screen, including both recovery criteria. On the primary grids, the best forecaster or imitator within 1% disagreement adds 4.39%/7.42%/4.92% bytes (Table VI), so self-triggering raises the attainable saving at that limit by 12–17 percentage points. Code, the frozen protocol, and the subject-equal summaries are in the online repository.

A. Learned Auxiliary Detector (V31)

After the V30 results were seen, a second variant was designed and frozen before its own A evaluation; it uses the same subjects. Each auxiliary replaces the fixed max-change rule with a logistic detector. Its inputs are the standardized level, normalized change since the last push, the absolute change, the largest absolute change, and the age, computed from the per-axis mean, standard deviation, and mean absolute first difference of its own window. The detector is trained on B only, with deterministic pairs of windows (ages 1–15, at most 6,000 per subject, subject-equal weights). Its label marks when this auxiliary's stale scores alone change the fused decision. The auxiliary pushes when the predicted probability exceeds θ, at age sixteen, or at a reset; the float32 detector package (about 0.5–0.9 kB per session) is charged. The deployable θ is chosen on B from $\{0.01, 0.02, 0.03, 0.05, 0.075, 0.1, 0.15, 0.2, 0.3\}$ with the B-measurable gates and a stricter 0.75% B disagreement margin, else the smallest value.

TABLE VIII
Learned auxiliary detector. Frozen B-selected θ: byte saving (%), reference disagreement (%), calls/window. Last column: highest saving among the nine fixed θ with at most 1% disagreement, with disagreement and θ.

Dataset	Grid	Saving	Disagr.	Calls	Best within 1%
PAMAP2	all	10.74	1.36	2.639	6.22 (0.64; $\theta = 0.01$)
PAMAP2	alt 0	6.95	0.95	2.742	5.16 (0.64; $\theta = 0.01$)
PAMAP2	alt 1	6.91	1.06	2.742	5.47 (0.78; $\theta = 0.01$)
MHEALTH	all	5.54	0.47	2.794	10.36 (0.93; $\theta = 0.03$)
MHEALTH	alt 0	3.35	0.44	2.823	7.40 (0.83; $\theta = 0.03$)
MHEALTH	alt 1	2.97	0.37	2.834	7.29 (0.83; $\theta = 0.03$)
OPPORTUNITY	all	-0.06	0.07	2.994	5.10 (0.67; $\theta = 0.03$)
OPPORTUNITY	alt 0	-3.18	0.00	3.000	0.76 (0.93; $\theta = 0.05$)
OPPORTUNITY	alt 1	-3.16	0.00	3.000	-1.27 (0.71; $\theta = 0.05$)

No B-selected setting passes the screen in any scenario (Table VIII). Selection often falls back to near-complete push, especially on OPPORTUNITY, where each case has a single B subject. Within 1% disagreement, the detector improves on the fixed rule on the MHEALTH primary grid, where the fixed $\theta = 0.03$ passes every gate, and reaches the 1% limit on the PAMAP2 nonoverlap grids and OPPORTUNITY alt 1, where the fixed rule does not. It is worse on the PAMAP2 primary grid, the MHEALTH nonoverlap grids, and the other two OPPORTUNITY grids. A detector fitted on one to four subjects therefore does not transfer reliably; more training subjects or richer auxiliary evidence would be needed.

B. Two-Sided Trigger (V32)

The third design, frozen after the V30 and V31 results were seen and before its own A evaluation, keeps the V30 auxiliary rule and adds a primary margin guard. After auxiliary pushes, the primary fuses its current scores with the latest auxiliary scores. If some auxiliary is stale and the gap between the two largest uint8 score sums is at most δ, the primary sends one 30-byte notice pair and every stale auxiliary infers and sends fresh scores. The package adds δ to the V30 trigger package. On B only, (κ, δ) is chosen from $\{1, 1.5, 2, 3, 4\} \times \{4, 8, 16, 24, 32, 48\}$ as the lowest-byte pair meeting the B-measurable gates at the unchanged 1% disagreement limit, and otherwise the most conservative pair ($\kappa = 1$, $\delta = 48$). The δ grid spans the 5th to 25th percentiles of fused margins in one external B frame; no outcome was inspected when it was set. A B-only self-test checks an independent replay, that a guard that never fires reproduces V30 exactly, and that a guard that always fires reproduces complete-push fusion.

TABLE IX
Two-sided trigger, frozen B selection. Byte saving (%), reference disagreement (%), calls/window, pulls (% of windows), auxiliary pushes including pulls (% of auxiliary-windows), known-endpoint macro-F1 loss (percentage points), and screen result.

Dataset	Grid	Saving	Disagr.	Calls	Pulls	Pushes	F1 loss	Screen
PAMAP2	all	12.40	1.02	2.459	9.1	73.0	-0.18	disagreement
PAMAP2	alt 0	7.03	0.71	2.575	12.0	78.7	-0.13	saving
PAMAP2	alt 1	6.53	0.71	2.594	11.7	79.7	-0.16	saving
MHEALTH	all	16.55	0.77	2.292	11.1	64.6	-0.06	pass
MHEALTH	alt 0	13.12	0.75	2.332	13.6	66.6	-0.07	pass
MHEALTH	alt 1	13.43	0.74	2.322	13.7	66.1	-0.21	pass
OPPORTUNITY	all	18.08	0.61	2.231	14.0	61.5	0.11	pass
OPPORTUNITY	alt 0	5.69	0.52	2.650	6.2	82.5	0.00	saving
OPPORTUNITY	alt 1	7.47	0.72	2.592	6.5	79.6	-0.25	saving

Every evaluation subject saves bytes in all nine scenarios, and both recovery criteria pass everywhere (Table IX). The B selection varied across case–grids; the most frequent pairs were $(\kappa, \delta) = (1, 48)$, $(4, 24)$, $(3, 24)$, and $(3, 16)$. Among the 30 fixed pairs, none passes all six original or all three external scenarios; the best pass four scenarios each. On the PAMAP2 primary grid the fixed pair $(4, 24)$ passes with 10.63% saving at 0.75% disagreement, but it was not the B-selected rule and is not claimed. Within 1% disagreement, no fixed pair reaches 10% saving on the PAMAP2 or OPPORTUNITY nonoverlap grids (best 7.85%–9.32%).

VII. Cost-Aware Design Selection

Scope. Audit cost screening applies the exact cost estimate of the main paper's Eq. (7); it reproduces ordinary serialized cost search and is not a new optimizer. The ten-byte control pair sends a uint32 global window identifier and a uint8 action to each auxiliary. It removes the shared session-local counter assumed by the earlier six-byte format but still assumes a shared window stream, bound reliable ordered delivery, and nonwrapping identifiers. Three 18-byte schema messages add 54 bytes per session to each changed implementation and to its matched complete-push baseline. In each cyclic role, B-fit uses ADL1–3 and B-select uses ADL4–5, so no landmark spans fitting and selection; node models and the current-risk head stay frozen, and all comparators share one isotonic calibrator. Exhaustive cost search and audit screening share the 24 candidates, every quality and resource gate, the objective, and the tie order. Fixed-width and matched-width 0.01 controls separate setting selection from packing, and minimum-call tuning ignores bytes but keeps the quality and call constraints. Serialization counts exclude shared schedule generation and quality scoring, so no end-to-end speedup is claimed. The A results are development evaluations of previously exposed subjects; because the prespecified condition was not met, the held-out Drill sessions were not opened.

Protocol. OPPORTUNITY uses one-based accelerometer columns 38–40 (BACK), 64–66 (RLA), and 90–92 (LLA), timestamp column one in milliseconds, and locomotion-label column 244. Labels 1/2/4/5 map to slots 0/1/2/3; other labels are unknown. Each of twelve cyclic subject/grid cases fits isotonic calibration on up to 512 B-fit landmarks with the frozen 52-feature current-risk head. The seven thresholds are 0.005, 0.01, 0.02, 0.03, 0.05, 0.075 and 0.1; maximum interval remains sixteen. Four periodic controls use 2/4/8/16; complete push is the fallback. Twelve/four-slot representations yield twenty-four candidates. Risk thresholds are encoded as float32. The candidate order is ascending thresholds, ascending periods, then complete push, with twelve slots preceding four for each rule.

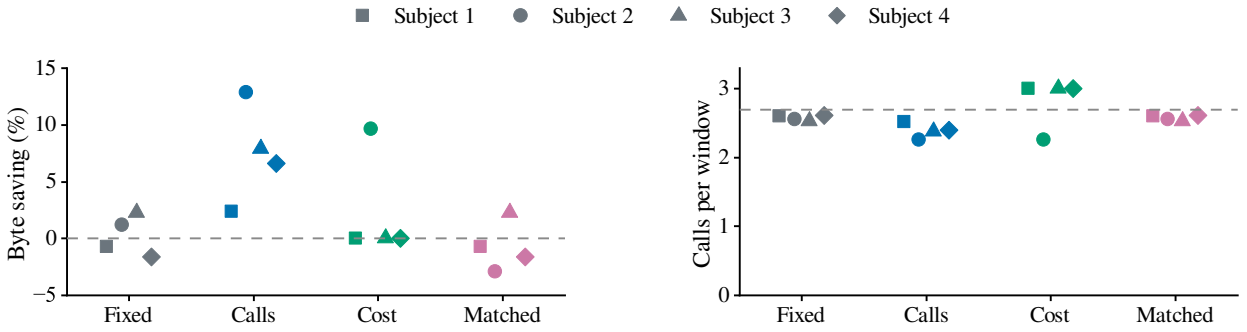
Quality constraints include disagreement, known-endpoint F1 loss, first-correct and two-consecutive-correct recovery. Added failure is checked separately from p95 added windows among pairs that both recover. Resource constraints are ten percent mean-byte saving, positive savings for at least seventy-five percent of subjects, and at most 2.7 calls/window. B-select has one person per case, so its positive-subject proportion is either zero or one. Selection minimizes absolute application bytes per window among eligible candidates, breaking ties by candidate order. Empty sets produce complete twelve-slot push, explicitly marked as fallback rather than success. All twelve choices were durably saved before new A scoring. A subject is excluded from both B calibration and S model fitting, but ADL outcomes had prior development exposure. The sixty selected A executions and twenty conditional raw-window executions match causal schedules and decoded packet ledgers. Both width-specific complete baselines are rerun in each case. Every score/control packet is parsed. Explicit control rejects malformed, invalid-action, duplicate or stale window identities.

Results. Table X gives the aggregate primary-grid results; the tables below list every B-side choice and every A evaluation setting, and the figure shows each subject.

TABLE X

Development primary-grid results with shared calibration. Saving compares subject-equal mean bytes against complete push in each selected width. Matched fixes 0.01 at the audit width. Failed gates: B, bytes; S, positive subjects; C, calls; D, disagreement; F, F1; R, recovery. F1 loss is in percentage points.

Selection	Saving (%)	Calls	D (%)	F1 loss (pp)	Positive	Failed gates
Fixed 0.01 / 12	0.28	2.5748	0.168	0.056	2/4	B,S
Minimum-call tuning	7.44	2.3916	0.425	0.158	4/4	B
Exhaustive cost search	2.04	2.8156	0.151	0.083	1/4	B,S,C
Audit cost screening	2.04	2.8156	0.151	0.083	1/4	B,S,C
Fixed 0.01 / matched	-0.64	2.5748	0.168	0.056	1/4	B,S



(a) Communication saving

(b) Classifier calls

Fig. 1. Four OPPORTUNITY subjects on the development primary grid. Fixed uses isotonic 0.01/twelve slots; Calls minimizes calls subject to quality and call constraints; Cost is audit cost screening, identical to exhaustive cost search; Matched fixes 0.01 at Cost's selected width. Each subject's byte saving uses complete push in the same selected wire format. Symbols identify individuals; horizontal offsets separate points. The zero-saving line and the 2.7-call aggregate limit provide reference levels. The latter is a condition on the subject-equal mean, not an individual requirement. Table X reports the complete aggregate screen.

TABLE XI: B-side choices and required full serialization evaluations. Push denotes no eligible candidate.

Case/grid	Audit choice	Feasible	Minimum-call choice	Exhaustive	Audit
A1 1/0	push / 12	0	iso 0.03 / 12	9	1
A1 2/0	push / 12	0	push / 12	1	1
A1 2/1	push / 12	0	push / 12	1	1
A2 1/0	iso 0.075 / 4	8	iso 0.075 / 12	14	1
A2 2/0	push / 12	0	push / 12	1	1
A2 2/1	push / 12	0	iso 0.1 / 12	5	1
A3 1/0	push / 12	0	iso 0.05 / 12	11	1
A3 2/0	push / 12	0	push / 12	1	1
A3 2/1	push / 12	0	iso 0.05 / 12	5	1
A4 1/0	push / 12	0	iso 0.05 / 12	11	1
A4 2/0	push / 12	0	push / 12	1	1
A4 2/1	push / 12	0	iso 0.1 / 12	5	1

TABLE XII: All new A evaluation settings. P denotes the number of subjects with byte saving; fallback counts are specific to the listed selector. F is known-endpoint F1 loss in percentage points.

Grid	Selector	Saving (%)	Calls	D (%)	F (pp)	P; fallback	Failed
1/0	Fixed 0.01 / 12	0.277	2.5748	0.168	0.056	2/4; 0	B,S
1/0	Minimum-call tuning	7.443	2.3916	0.425	0.158	4/4; 0	B
1/0	Exhaustive cost search	2.035	2.8156	0.151	0.083	1/4; 3	B,S,C
1/0	Audit cost screening	2.035	2.8156	0.151	0.083	1/4; 3	B,S,C
1/0	Fixed 0.01 / matched	-0.640	2.5748	0.168	0.056	1/4; 0	B,S
2/0	Fixed 0.01 / 12	-13.623	2.8800	0.235	0.002	0/4; 0	B,S,C
2/0	Minimum-call tuning	0.000	3.0000	0.000	0.000	0/4; 4	B,S,C
2/0	Exhaustive cost search	0.000	3.0000	0.000	0.000	0/4; 4	B,S,C
2/0	Audit cost screening	0.000	3.0000	0.000	0.000	0/4; 4	B,S,C
2/0	Fixed 0.01 / matched	-13.623	2.8800	0.235	0.002	0/4; 0	B,S,C

Grid	Selector	Saving (%)	Calls	D (%)	F (pp)	P; fallback	Failed
2/1	Fixed 0.01 / 12	-15.203	2.9192	0.115	0.002	0/4; 0	B,S,C
2/1	Minimum-call tuning	-2.298	2.7082	0.736	-0.059	0/4; 1	B,S,C
2/1	Exhaustive cost search	0.000	3.0000	0.000	0.000	0/4; 4	B,S,C
2/1	Audit cost screening	0.000	3.0000	0.000	0.000	0/4; 4	B,S,C
2/1	Fixed 0.01 / matched	-15.203	2.9192	0.115	0.002	0/4; 0	B,S,C

The primary-grid exhaustive/audit denominator averages the complete-push cost in each subject's B-selected width; three subjects use twelve slots after fallback and one uses four. Fixed-matched uses those same widths and the same new calibrators. Absolute policy and reference bytes are retained below to prevent conflating width changes and relative savings.

Primary-grid selector	Policy B/window	Matched push B/window	Twelve-slot push B/window
Fixed 0.01 / 12	66.177789	66.361453	66.361453
Minimum-call tuning	61.422304	66.361453	66.361453
Exhaustive cost search	61.903173	63.189214	66.361453
Audit cost screening	61.903173	63.189214	66.361453
Fixed 0.01 / matched	63.593524	63.189214	66.361453

VIII. Positioning against Decision-Focused Learning

Elmachtoub and Grigas, Smart Predict, then Optimize (Management Science 68(1), 9–26, 2022; DOI 10.1287/mnsc.2020.3922), use optimization structure to train prediction models. Mandi et al., Decision-Focused Learning: Foundations, State of the Art, Benchmark and Future Opportunities (J. Artif. Intell. Res. 80, 1623–1701, 2024; DOI 10.1613/jair.1.15320), survey methods that train predictors through downstream decisions. These establish that predictive accuracy and decision utility can diverge. Our supported distinction is the matched diagnosis of frozen forecasts, action regions, serialized control charges and remaining zero-conflict scheduling space. We do not claim to originate prediction–decision misalignment or to outperform decision-focused training.

IX. Reproducibility

The online repository provides the offline-schedule and imitation code, the frozen imitation protocol with its source hash, and the subject-equal summary tables behind the main paper's numbers and Fig. 3. The full experiment bundle additionally retains all B candidate schedules, frozen choices, A and raw-prefix traces, independent byte and quality audits, input and source hash locks, calibrator packages, and score frames; re-executing original inference requires the raw datasets and fitted node models. Independent checks reconstruct 288 B candidate costs, sixty A streams, twenty raw-prefix paths, and fifteen weighted summaries, and a separate implementation verifies all 288 B quality records and sixty A outcomes using exact rational weights for the p95 recovery mixtures. Figs. 2 and 3 of the main paper are drawn from `figure_envelope_points.csv`, `frontier_points.csv`, the self-trigger summaries `aux_trigger_v30_points.csv` and `aux_trigger_v30b_points.csv`, and `twoside_trigger_v32_points.csv`; the subject-level design figure here is an Origin plot. No new individuals, radio measurements, or physical-network timings are introduced.